UNIVERSITY
ROORKEE, UTTARAKHAND

7th, KM Haridwar, National Highway Vardhmanpuram,
Roorkee, Rehmadpur, Uttarakhand 247667

Lab Sheet-05

Python Programming Lab – Tuples Operations

MCA I Semester (Session 2026-2027)

Objectives:

- To understand the use of tuples for storing and managing data in Python.
- To implement basic and advanced operations on tuples for solving real-life problems.

Requirements:

- Python 3.x installed on the system (IDLE, PyCharm, VS Code, or any preferred IDE).
- Basic knowledge of Python data types, especially **tuples**.
- Understanding of input/output operations, indexing, slicing, and iteration.

Program List:

1. Write a Python program to create and display a tuple.
2. Write a program to create a tuple with different data types.
3. Write a program to find the length of a tuple.
4. Write a program to access tuple elements using indexing.
5. Write a program to access tuple elements using slicing.
6. Write a program to check if an element exists in a tuple.
7. Write a program to count the occurrences of an element in a tuple.
8. Write a program to find the index of an element in a tuple.
9. Write a program to concatenate two tuples.
10. Write a program to repeat a tuple n times
11. Write a program to convert a list into a tuple.
12. Write a program to convert a tuple into a string.
13. Write a program to find the maximum and minimum values in a tuple.

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14. Write a program to calculate the sum of numeric elements in a tuple.
15. Write a program to sort a tuple of numbers in ascending order.
16. Write a program to sort a tuple of strings alphabetically.
17. Write a program to reverse a tuple.
18. Write a program to check if two tuples are equal.
19. Write a program to remove duplicate elements from a tuple.
20. Write a program to find the length of the longest word in a tuple.
21. Write a program to create a nested tuple and access its elements.
22. Write a program to flatten a nested tuple.
23. Write a program to swap two tuples.
24. Write a program to find common elements between two tuples.
25. Write a program to find the difference between two tuples.
26. Write a program to convert a tuple of tuples into a dictionary.
27. Write a program to unzip a list of tuples into individual lists.
28. Write a program to find repeated items in a tuple.
29. Write a program to check if all elements in a tuple are the same.
30. Write a program to find the element-wise sum of two tuples.

Instruction:

- Write each program in a separate .py file with proper naming (e.g., Q1_tuple_creation.py).
- Use **user input** wherever applicable.
- Provide **comments** inside code to explain logic.
- Test each program with different inputs before submission.
- Display output in a **clear and readable format**.

Guideline:

- Do not use **advanced Python libraries** unless specified.
- Focus on **tuple operations only** (avoid unnecessary conversion to lists unless asked).

- Follow proper **Python indentation (4 spaces)**.
- Use meaningful variable names.
- Ensure programs run without syntax or runtime errors.

Submission:

- Submit the source code files (.py) for each task along with a brief report documenting the implementation details and the results of the executions.



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